Yuanqing Ma

Shenzhen, China | yuanqingma@link.cuhk.edu.cn | (+86) 13612923691 | Tommy Ma | Homepage

Education

BS **The Chinese University of Hong Kong, Shenzhen**, Computer Science and Engineering — Sept 2024 - Present

- **CGPA**: 3.929/4.000
- **CGPA Ranking in Major**: **5**/238
- **Coursework:** Honors Calculus I & II, Linear Algebra, Data Structures, Matrix Computations, Honors Machine Learning, Reinforcement Learning

Research Experience

Research Assistant, *The Chinese University of Hong Kong, Shenzhen* — Sept 2025 - Present

Research Topic: In-Context Reinforcement Learning

Advised by Prof. Chenjun Xiao (School of Data Science, CUHK(SZ))

- **Algorithm Reproduction:** Implemented the *Algorithm Distillation (AD)* framework, distilling RL algorithms into Transformers and validating zero-shot generalization ability in the Darkroom environment (code available on GitHub).
- **Method Innovation:** Addressed the long-context inference bottlenecks in AD by designing a context compression mechanism, reducing the context window to 20%–25% of the original size while maintaining performance and significantly increasing inference speed.
- **Detailed Implementation:** Used multiprocessing to efficiently obtain PPO training history with parallel threads; implemented Transformer architectures in PyTorch and conducted large-scale experiments on Grid-world and Meta-world tasks.
- **Research Output:** First-author paper under review in ICML 2026.

Projects

Autonomous AI Data Scientist Agent — GitHub

- Designed an end-to-end agentic pipeline that automates the ML lifecycle (EDA to evaluation) using local LLMs.
- Developed a self-healing code execution framework that drafts, debugs via traceback analysis, and iteratively improves the code in a sandboxed environment.
- Leveraged custom logging to track and export top-performing model artifacts.

MNIST Image Classification via Contextual Bandits — GitHub

- Reformulated supervised learning as a multi-armed bandit problem using Gymnasium and self-designed reward function.
- Applied the REINFORCE algorithm to optimize the policy network. Incorporated entropy regularization to stabilize training.

Course Projects

Reinforcement Learning — Fall 2025

- Implementation of DQN on the game of Cartpole: Colab
- Implementation of PPO on the Pendulum-v1 Environment: Colab
- Literature Review on In-context Reinforcement Learning: PDF

Computational Laboratory Using Java Spring 2025

https://github.com/tommyma3/CSC1004

- **Java "Chatroom"**: Supports multiple threads and command-line interfaces, allowing each user to send and receive messages to and from all others in real-time
- **Gomoku Game**: Traditional "Gomoku" game with JavaFX GUI
- **Python Snake Game**: Traditional Snake Game implemented with Python

Teaching

Undergraduate Student Teaching Fellow (USTF), CUHK(SZ)

- CSC1003: **Introduction to Computer Science and Java Programming**, Fall 2025
- MAT2041A: **Foundations of Linear Algebra and its Applications**, Spring 2026

Honors & Scholarships

Dean's List Award, School of Data Science, CUHK(SZ) AY2024-25

Academic Performance Scholarship: Class A, School of Data Science, CUHK(SZ) AT2024-25

Master's List, Shaw College, CUHK(SZ) AY2024-25

Full-tuition Scholarship & Sin Wai Kin Special Admissions Scholarship, CUHK(SZ) Sept 2024 - Present

Skills

Programming Languages: C, C++, Java, Python, RISC-V

Tools: Git, GitHub, PyTorch, NumPy, LaTeX, Markdown

Language Proficiency: English

Extracurricular Activity

Shaw College Men's Soccer Team Sept 2024 - Present

- Player & Assistant Coach
- Won the *Shaw Outstanding Student Organization Award* of AY 2024-25.
- Won the Gold Medal in the Men's Soccer Competition at the CUHK(SZ) Olympics in April 2025

Student Assistant for the Information Technology Service Office (ITSO) at CUHK(SZ) Oct 2024 - Present

- Technical Assistant
- Assisting ITSO staff in offering IT service on campus, including the CUHK(SZ) email and websites